# EXPERIMENT 8

## Report Scheduling, Automation and Alerts using Microsoft Power BI

### AIM

To design and implement report scheduling, automation, and alert mechanisms using Microsoft Power BI and develop strategies for automated report delivery, monitoring, and business performance tracking.

### TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel

### THEORY

Business Intelligence systems must deliver information to stakeholders at the right time. Scheduled refresh keeps the dataset current, subscriptions e-mail snapshots of report pages on a fixed frequency, and data-driven alerts notify users when a KPI tile crosses a threshold. Together they move reporting from 'pull' (user opens report) to 'push' (insight reaches the user).

### BUSINESS SCENARIO

Global Superstore management requires automated delivery of reports and timely notifications whenever business performance changes significantly. As BI Operations Engineer, a reporting automation strategy is designed.

### DATASET USED

Global_Superstore_2018.xlsx - sheets Orders (51,290 rows, 2014-2017), Returns (2,033 rows) and People (24 rows). Overall: Sales $12,642,502, Profit $1,467,457, Profit Margin 11.6%, 25,728 orders, 1,970 returned orders.

### PROCEDURE

- Load Orders, Returns, People into Power BI Desktop (Home > Get Data > Excel).
- Create DAX measures: Total Sales, Total Profit, Profit Margin, Sales YoY %, Returned Orders, alert flags.
- Build the 3-page dashboard (Executive Overview, Regional Analysis, Alerts and Monitoring).
- Publish to Power BI Service workspace 'Superstore BI Ops' (Home > Publish).
- Dataset > Settings > Scheduled refresh: set daily refresh time and credentials.
- Report > Subscribe to report: create the Daily / Weekly / Monthly subscriptions of Part A.
- Pin KPI cards to a dashboard; tile '...' > Manage alerts for each rule of Part C.

- Optional: Power Automate flow to send a Teams/e-mail message and create a task when an alert fires.

## CONDUCT - PART A: SCHEDULED REPORTING PLAN

| Report | Content | Intended Users | Frequency | Schedule (IST) | Dataset Refresh |
|---|---|---|---|---|---|
| **Daily Sales Report** | Sales, orders, AOV, returns by region | Sales Managers, Regional Managers | Daily | 07:00 Mon-Sat | 05:30 daily |
| **Weekly Profit Report** | Profit, margin, discount impact by category/region | Finance Team, Regional Heads | Weekly | Monday 08:00 | Sunday 23:00 |
| **Monthly Executive Dashboard** | Sales vs PY, margin trend, return rate, alert summary | CEO, CFO, Directors | Monthly | 1st of month 09:00 | Last day 23:30 |

## CONDUCT - PART B: REPORT DISTRIBUTION MATRIX

| User Group | Report | Frequency | Delivery Method |
|---|---|---|---|
| **Sales Managers** | Daily Sales Report | Daily | Email subscription (image/link) |
| **Regional Managers** | Daily Sales Report | Daily | Email + Power BI Service app |
| **Finance Team** | Weekly Profit Report | Weekly | Email with PDF/PowerPoint attachment |
| **Regional Heads** | Weekly Profit Report | Weekly | Shared Workspace |
| **Executives (CEO, CFO)** | Monthly Executive Dashboard | Monthly | Email PDF + Power BI Service |
| **BI/IT Operations** | Refresh failure & usage metrics | On event | Email alert / Service notification |

## CONDUCT - PART C: KPI ALERT CONFIGURATION SHEET

| KPI | Alert Condition | Check | Recipients | Action |
|---|---|---|---|---|
| **Profit Margin** | Below 10% | Daily (Service alert) | Regional Manager, Finance Head | Review discounts and costs; control extra discounting |
| **Sales (YoY/MoM)** | Decline greater than 15% | Daily | Sales Head, Regional Manager | Promotion/customer outreach; root-cause by product |

| | | | | |
|---|---|---|---|---|
| **Returned Orders** | Increase greater than 20% | Daily | Operations Head, Quality Team | Investigate product/vendor/delivery issue |
| **Dataset Refresh** | Refresh failed | On failure | BI Operations | Fix credentials/gateway and re-run |

Power BI note: data alerts work on card/KPI/gauge tiles pinned to a dashboard. Percentage rules are implemented as DAX flag measures (e.g. Margin Alert) or by setting the alert threshold directly on the Profit Margin card (e.g. below 0.10).

## DAX MEASURES USED

- Total Sales = SUM(Orders[Sales])
- Total Profit = SUM(Orders[Profit])
- Profit Margin = DIVIDE([Total Profit],[Total Sales])
- Sales PY = CALCULATE([Total Sales], REMOVEFILTERS(Orders[Year]), Orders[Year] = MAX(Orders[Year]) - 1)
- Sales YoY % = DIVIDE([Total Sales]-[Sales PY],[Sales PY])
- Returned Orders = CALCULATE(DISTINCTCOUNT(Orders[Order ID]), TREATAS(VALUES(Returns[Order ID]), Orders[Order ID]))
- Margin Alert = IF([Profit Margin] < 0.10, "ALERT", "OK")
- Sales Decline Alert = IF([Sales YoY %] < -0.15, "ALERT", "OK")

## DASHBOARD SCREENSHOT

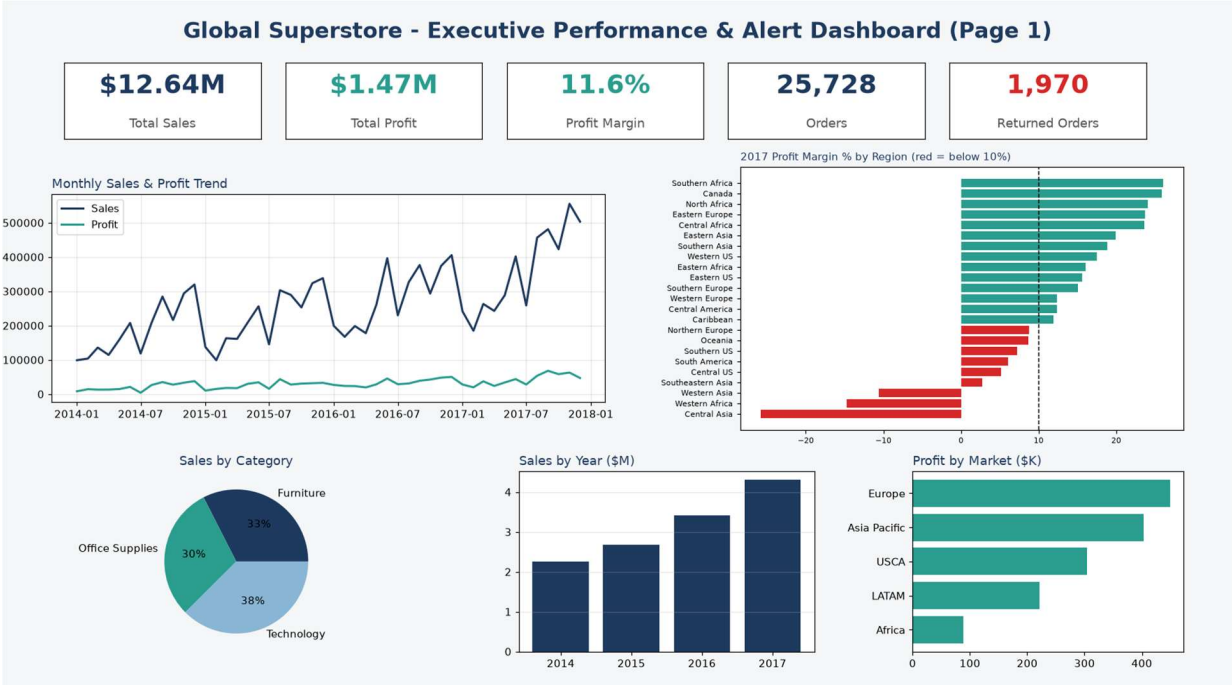
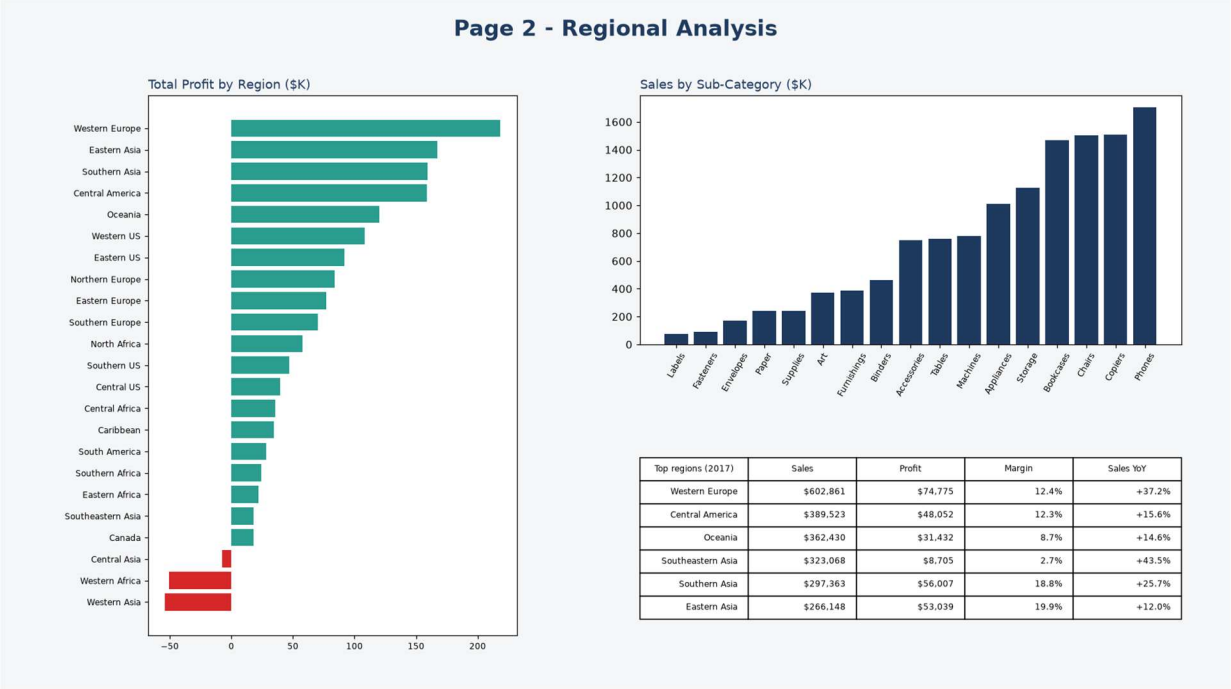
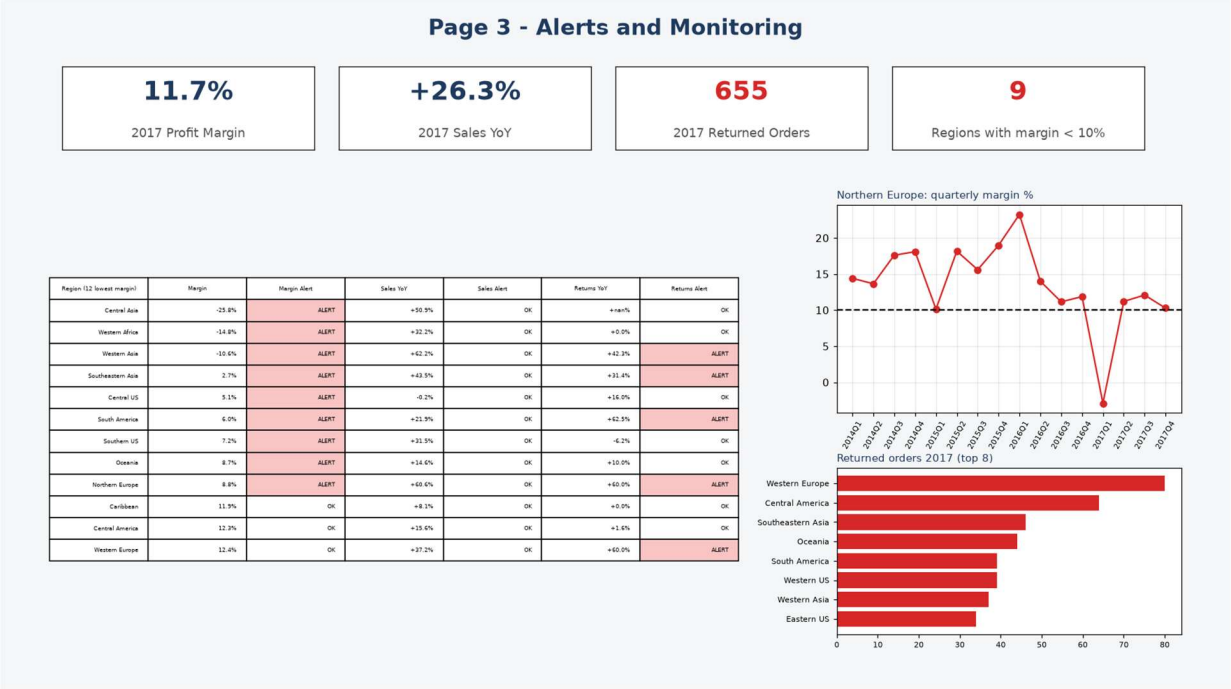


Figure 1: Page 1 - Executive Overview (preview of the dashboard; replace with Power BI Desktop screenshot).




| Top regions (2017) | Sales | Profit | Margin | Sales YoY |
|---|---|---|---|---|
| Western Europe | $602,861 | $74,775 | 12.4% | +37.2% |
| Central America | $389,523 | $48,052 | 12.3% | +15.6% |
| Oceania | $362,430 | $31,432 | 8.7% | +14.6% |
| Southeastern Asia | $323,068 | $8,705 | 2.7% | +43.5% |
| Southern Asia | $297,363 | $56,007 | 18.8% | +25.7% |
| Eastern Asia | $266,148 | $53,039 | 19.9% | +12.0% |

Figure 2: Page 2 - Regional Analysis.




| Region (12 lowest margin) | Margin | Margin Alert | Sales YoY | Sales Alert | Returns YoY | Returns Alert |
|---|---|---|---|---|---|---|
| Central Asia | -25.8% | ALERT | +50.5% | OK | +nan% | OK |
| Western Africa | -14.8% | ALERT | +32.2% | OK | +0.0% | OK |
| Western Asia | -10.6% | ALERT | +62.2% | OK | +42.3% | ALERT |
| Southeastern Asia | 2.7% | ALERT | +43.5% | OK | +31.4% | ALERT |
| Central US | 5.1% | ALERT | -0.2% | OK | +16.0% | OK |
| South America | 6.0% | ALERT | +21.9% | OK | +62.5% | ALERT |
| Southern US | 7.2% | ALERT | +31.5% | OK | -6.2% | OK |
| Oceania | 8.7% | ALERT | +14.6% | OK | +10.0% | OK |
| Northern Europe | 8.8% | ALERT | +69.6% | OK | +60.0% | ALERT |
| Caribbean | 11.9% | OK | +8.1% | OK | +0.0% | OK |
| Central America | 12.3% | OK | +15.6% | OK | +1.6% | OK |
| Western Europe | 12.4% | OK | +37.2% | OK | +60.0% | ALERT |

Figure 3: Page 3 - Alerts and Monitoring.

## CONDUCT - PART D: MONITORING SCENARIO ANALYSIS

Scenario: the Northern Europe regional branch shows declining profitability. Its profit margin fell from 15.3% (2016) to 8.8% (2017), below the 10% threshold. Sales changed +60.6% YoY and returned orders changed +60.0% YoY. Average discount moved from 16.3% to 15.4%. Weakest sub-categories in 2017: Tables ($-3,804), Furnishings ($-1,547), Supplies ($29).

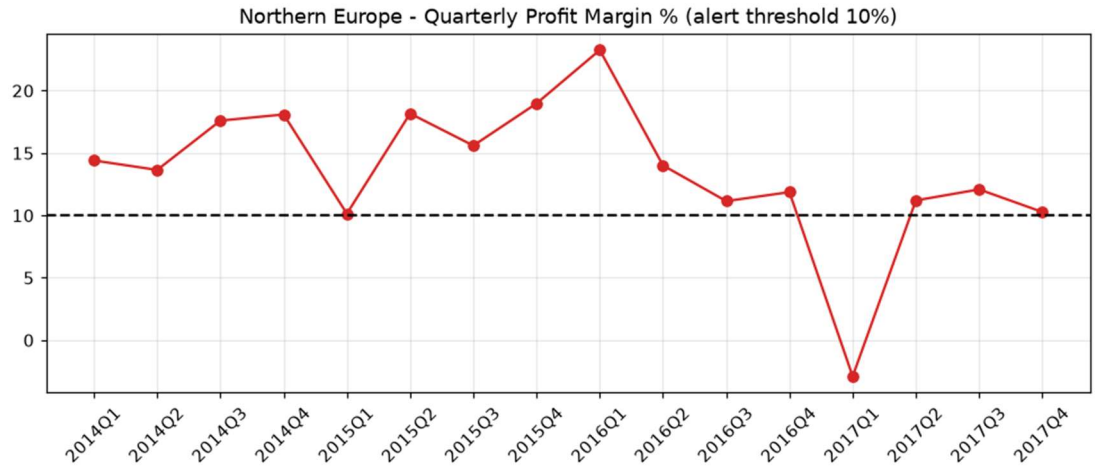



Figure 4: Northern Europe quarterly profit margin vs 10% alert threshold.

| Alert Trigger | Observed | Recipients | Corrective Action |
|---|---|---|---|
| **Profit Margin < 10%** | Northern Europe 8.8% | Northern Europe Regional Manager, Finance Head | Review discounts, cap deep-discount orders, check shipping cost |
| **Sales decline > 15%** | +60.6% (not triggered) | Sales Head | Monitor; run targeted campaigns if decline continues |
| **Returns increase > 20%** | +60.0% (triggered) | Operations Head | Audit quality/logistics for returned orders |

**Regions below 10% margin in 2017**

| Region | Sales | Profit | Margin | Sales YoY | Returns YoY |
|---|---|---|---|---|---|
| **Central Asia** | $7,873 | $-2,033 | -25.8% | +50.9% | +nan% |
| **Western Africa** | $66,004 | $-9,742 | -14.8% | +32.2% | +0.0% |
| **Western Asia** | $115,902 | $-12,305 | -10.6% | +62.2% | +42.3% |
| **Southeastern Asia** | $323,068 | $8,705 | 2.7% | +43.5% | +31.4% |
| **Central US** | $147,098 | $7,551 | 5.1% | -0.2% | +16.0% |
| **South America** | $211,618 | $12,751 | 6.0% | +21.9% | +62.5% |

| | | | | | |
|---|---|---|---|---|---|
| **Southern US** | $122,977 | $8,861 | 7.2% | +31.5% | -6.2% |
| **Oceania** | $362,430 | $31,432 | 8.7% | +14.6% | +10.0% |
| **Northern Europe** | $237,433 | $20,827 | 8.8% | +60.6% | +60.0% |

## MONITORING AND RECOMMENDATION REPORT

- Switch on the margin alert for Northern Europe and escalate to the Regional Manager and Finance within 24 hours.
- Require approval for discounts above 20%; review loss-making sub-categories.
- 9 regions are below the 10% margin target in 2017; review them in the weekly profit report.
- Add return-reason capture and a weekly quality review for regions where returns grow more than 20%.
- Monitor dataset refresh with failure notifications to BI Operations.
- Review alert thresholds quarterly to avoid alert fatigue.

## RESULT

Report scheduling, automation, and alert strategies were successfully designed. Automated reporting plans and monitoring mechanisms were developed to support business operations.

## LEARNING OUTCOMES

- Applied report scheduling concepts in BI systems.
- Designed automated report distribution strategies.
- Analyzed KPI monitoring requirements.
- Evaluated alert mechanisms for business decision support.
- Developed recommendations for BI operations.